Saqi

Offline, decentralized water intelligence for the farms the cloud forgets

“The future of farm AI isn’t in the cloud. It’s in the field — with no internet at all.”

Fatma Ezzahra Jlali · Pioneer High School, Manouba, Tunisia · Moonshot 2026

THE BET

Everyone is trying to bring the cloud to the farm.
That’s backwards.

Cloud-first “smart farming” needs Wi-Fi, grid power, money, and data ; the exact things missing on the farms drought hits first. The tools meant to fight drought skip the people who need them most.
Saqi is the opposite: the intelligence lives in the field, on cheap devices that never touch a server.

THE PROBLEM

The farms the cloud forgets

500M+
smallholder farms worldwide

1/3
of the world's food they grow

1st
in line as the climate turns

Most are under two hectares, in the regions with the weakest infrastructure and the highest exposure to a warming climate.

Cloud-first design doesn't just underserve them ; it **structurally excludes** them, because its first requirement is the one thing they can't assume. And their real question is "who gets the last litre so nothing dies."

WHY CURRENT SYSTEMS FAIL

One brain, one link to break

The cloud

Every node reports to one gateway and waits for orders. When links or power drop, exactly when drought peaks, it goes blind and loses every zone it can no longer reach. A single point of failure, placed where infrastructure is weakest.

Saqi

No centre. Each node talks only to its neighbours and decides locally. Cut the network and it simply splits into smaller groves, each still keeping itself alive. Nothing to knock out.

THE IDEA

How Saqi works

1

A mesh of cheap, solar nodes

Each zone carries a few-dollar device with a soil probe and a short-range radio talking only to the zones beside it. No internet, no server.

2

A gossip “water-line”

Neighbours swap their water needs and average, over and over, until the whole patch agrees on one cutoff for where today’s scarce water should go with no central tally.

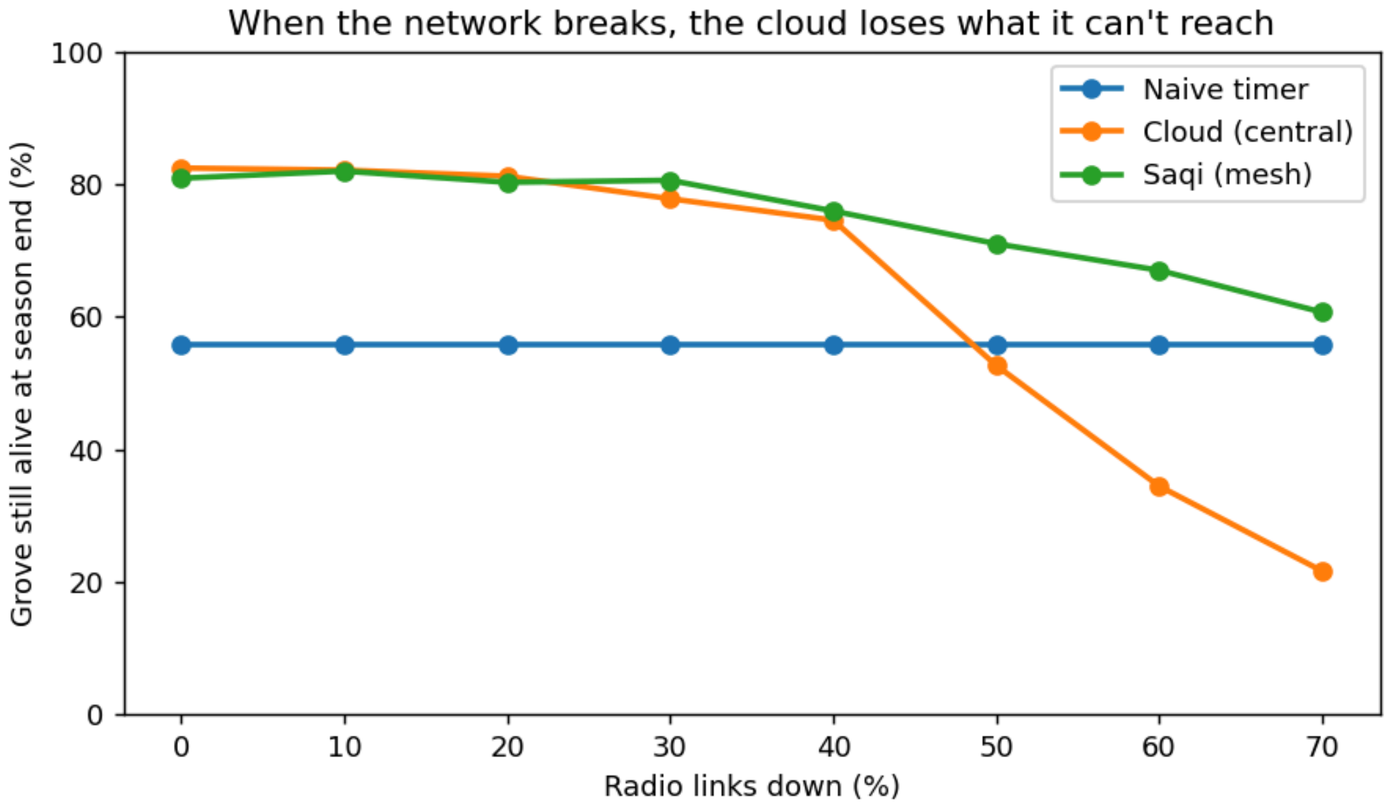
3

Keep every zone alive

Rescue the driest first, up to a thin survival line. Olives endure on little, so nothing strong is fattened while something weak dies.

THE RESULT THAT MATTERS

It matches the cloud until the cloud breaks



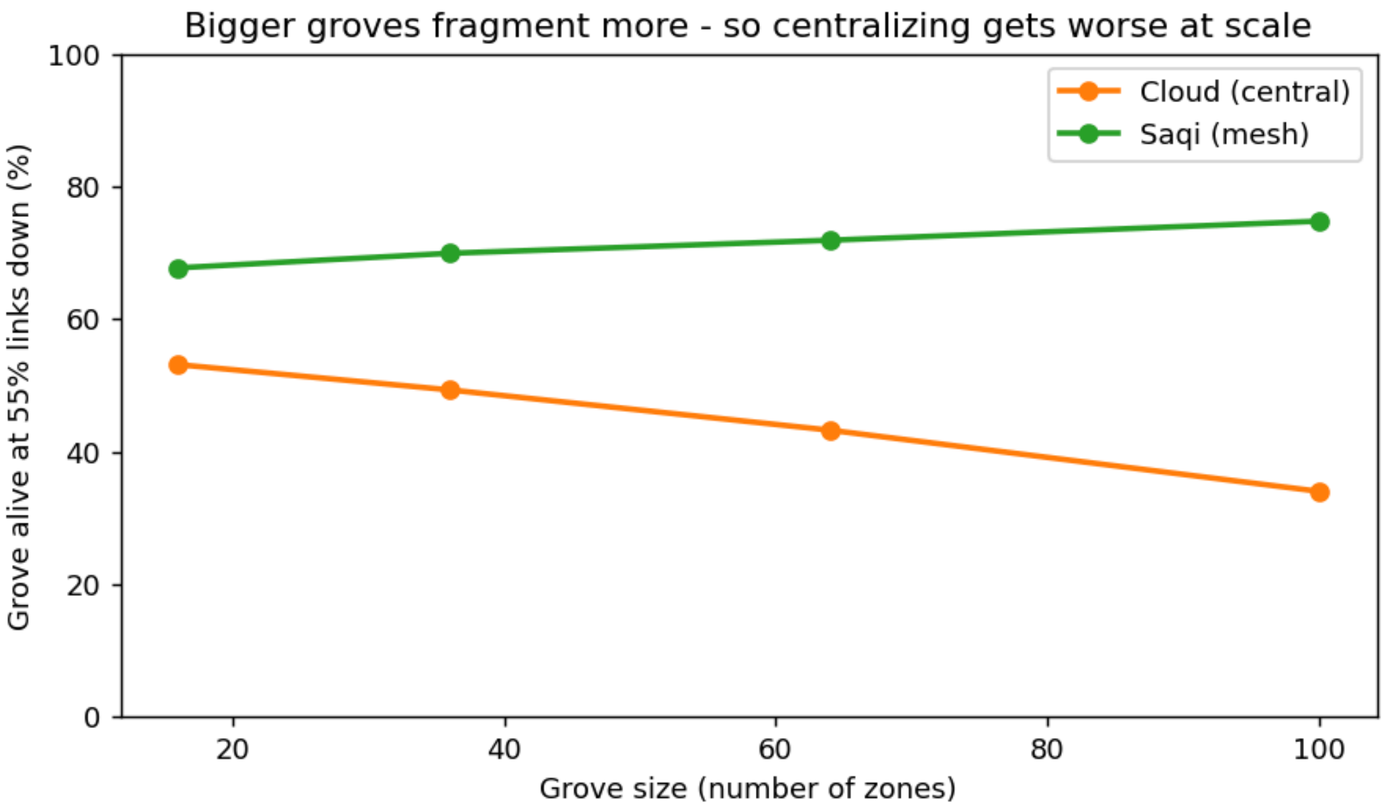
At 70% of links down

Cloud 22%
Saqi 61%

Smart but fragile (cloud) vs. robust but dumb (timer) vs. smart AND robust (Saqi).

AND IT SCALES THE RIGHT WAY

Bigger farms break centralization



In a degraded network, bigger groves fragment into more islands. So the cloud gets worse as the farm grows while Saqi holds and even improves.

Centralizing fails exactly where scale and stress meet.

WHY IT SHOULD EXIST

$20 a zone. No internet. No grid.

Precision farming, finally, for the half-billion farms that already feed much of the world and stand first in line as the climate turns against them.